

HCV Module for Viessmann Vitocal Heat Pumps

Description

HCV (Home Control Viessmann) is an adapter that integrates Viessmann heat pumps supporting E3 communication via CAN into a home control environment, without requiring advanced technical knowledge. In other words, apart from connecting the HCV module to the internal CAN bus and configuring Wi-Fi access, no further settings or expertise are required.

Vitocal 250	
DomesticHotWaterSensor	51.2
OutsideTemperatureSensor	17.8
MixerOneCircuitFlowTemperatur	25.5
WaterPressureSensor	1.7
PrimaryHeatExchangerLiquidTem	3.9
CentralHeatingPump	on
DomesticHotWaterPump	on
MixerOneCircuitPump	on
MixerOneCircuitRoomTemperatur	23.0
Time	16:48
EnergyConsumptionCentralHeati	53.6
MixerOneCircuitOperationState	on
PrimaryInletTemperatureSensor	17.7
SecondaryOutletTemperatureSen	26.8
PrimaryCircuitFanOne	32
PrimaryCircuitFanTwo	37
EvaporatorVaporTemperatureSen	7.9
CompressorSpeedPercent	20
HeatPumpCompressor	on
CurrentThermalCapacitySystem	3669

Once installed, data retrieved via the CAN bus can be viewed using any local web browser. If integration into a home control environment is not required, the installation process is complete at this stage. Current data can then be queried at any time by navigating to the address **hcv.local** in a browser.

Initial Installation and Reset

By default, the integrated Access Point mode is active, allowing the HCV module to be accessed via the address `192.168.1.1`. This state is indicated by a flashing red LED.



The network named "HCV" will appear in your mobile phone's network settings and can be selected. Once connected, you can enter and save your local network's credentials (SSID and password). To switch to standard Wi-Fi mode: disconnect the power supply (USB plug), flip switch 2 to the "OFF" position, and reconnect the power. Once the HCV module's status LED flashes green, the device can be accessed via the regular Wi-Fi network using the address `hcv.local`.

Note: Under the "Vitocal Data" menu item—which also serves as the HCV module's main page—the message "No device detected" is displayed if no valid CAN messages are detected. The display updates once at least one valid message is recognized. The messages in the "Vitocal 250" window are displayed in either German or the origin defined E3onCan messages.

In the event of an issue, the local network settings can be modified at any time by flipping switch 2 to the "ON" position and subsequently restarting the device.

MQTT Broker Configuration

The screenshot shows a web interface titled "Home Control Viessmann V:0.83". Below the title bar is a section labeled "MQTT Konfiguration". It contains four input fields: "Broker IP" with the value "111.222.33.4", "Broker Port" with the value "1883", "Broker User" with the value "User", and "Broker Password" with masked characters "*****". At the bottom of this section is a blue button labeled "speichern".

Once a connection has been successfully established, the broker's address—and, if applicable, the username and password—can be entered via the "MQTT Broker Config" menu option. Saving the configuration data triggers a restart of the HCV module.

As soon as CAN messages valid for a Vitocal heat pump are detected, they also appear as MQTT messages on the local network or within the MQTT broker.

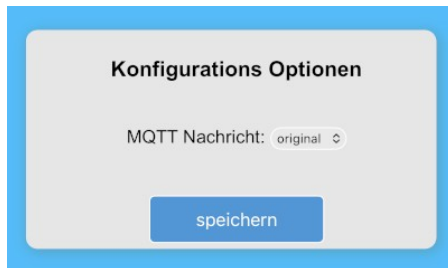
Note: The values (DIDs) currently defined in the Open3E datapoints are the ones transmitted.

We would like to take this opportunity to thank the creators of the document that served as the basis for the CAN messages for their work.

As previously mentioned, the CAN messages displayed in the browser are translated into German. The same setting can be applied to MQTT messages by selecting "German" under the "HCV Options" menu item. The advantage of the German version is that the MQTT messages are shorter.

MQTT Broker Message Format

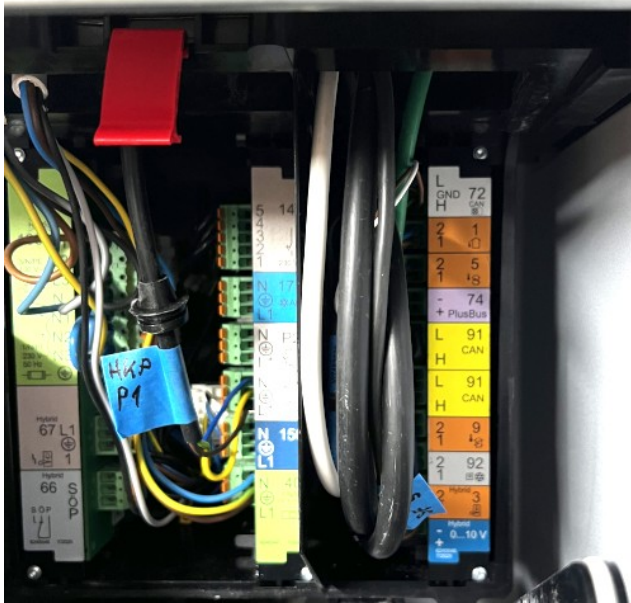
The MQTT messages displayed and sent by the HCV module can be customized.

The screenshot shows a web interface titled "Konfigurations Optionen". It contains a single dropdown menu labeled "MQTT Nachricht:" with the value "original" selected. At the bottom of this section is a blue button labeled "speichern".

Either the data IDs defined in the OpenE3datapoints document are used, or the German-language text strings are employed. This selection can be found under the "HCV Options" menu item.

CAN Bus Installation

The connection for detecting messages from Viessmann heat pumps is made via the internal CAN bus connector 72 of the Vitocal indoor unit. Since this connection is



already in use, the HCV module's sensor cable must be connected in parallel. Pay attention to the CAN bus polarity when doing so. The terminal on the HCV module corresponding to CAN High is marked in black.

Setting switch 1 to the "ON" position activates the CAN terminating resistor. This must be switched off when using the internal connection!

Note: The maximum length of the cable between the Vitocal connection and the HCV module connector must not exceed **10 meters**.

It should also be noted that the CAN connection is galvanically isolated to

prevent interference on both the CAN bus and the connected devices.

Status LED

LED flashes red: Access Point mode is selected via switch 2, or there is no connection to the configured local WLAN.

LED flashes red and green: Connection to the local WLAN established, but no CAN message detected.

LED flashes green: WLAN and CAN are active.

HCV Firmware Update

The HCV module's software can be updated via OTA (Over-the-Air), meaning directly over the network. Current versions are available on the embdes.com website and under [github /chrkir/hcv](https://github.com/chrkir/hcv). In the HCV menu, you will find this function under "Firmware update".



Support

For questions, errors, or suggestions, please write to info@embdes.com.

HCV MQTT ioBroker - Home Control – Node red

If the MQTT messages are to be integrated into a home automation application, this can be done by simply selecting the corresponding objects in the ioBroker MQTT tree. Other home control applications also accept this format; in some cases, the MQTT variant in JSON format is supported as well, offering the advantage of reduced system load.

File Explorer	Model Explorer	Model Explorer	Model Explorer
mqtt			
0			
1054	1054	folder	
EVC	EVC	state	variable
HCV	HCV	folder	
vcal	vcal	folder	
AllengraSensor	HCV/vcal/AllengraSensor	state	variable
BackendConnectionStatus	HCV/vcal/BackendConnect	state	variable
CentralHeatingPump	HCV/vcal/CentralHeatingP	state	variable
CompressorInletPressureSensor	HCV/vcal/CompressorInlet	state	variable
CompressorInletTemperatureSensor	HCV/vcal/CompressorInlet	state	variable
CompressorOilTemperatureSensor	HCV/vcal/CompressorOilTe	state	variable
CompressorOutletPressureSensor	HCV/vcal/CompressorOutle	state	variable
CompressorOutletTemperatureSensor	HCV/vcal/CompressorOutle	state	variable
CompressorSpeedPercent	HCV/vcal/CompressorSpec	state	variable
CurrentThermalCapacitySystem	HCV/vcal/CurrentThermalC	state	variable
DomesticHotWaterPump	HCV/vcal/DomesticHotWat	state	variable
DomesticHotWaterSensor	HCV/vcal/DomesticHotWat	state	variable
EconomizerLiquidTemperatureSensor	HCV/vcal/EconomizerLiqui	state	variable

HCV MQTT Home Assistant

As previously mentioned in the "Message Format" section, the HCV supports Home Assistant via a custom MQTT implementation. If the MQTT service is installed and enabled in Home Assistant, the Viessmann Vitocal heat pump is automatically detected and installed. Once the HCV module is connected to the internal CAN bus, automatic registration is initiated upon powering up the module. The values received from the control unit are then displayed in the Home Assistant "Devices" overview and continuously updated.

Viessmann Heat Pump >		Bereich zuordnen
	DomesticHotWaterSensor	51,2 °C
	OutsideTemperatureSensor	17,8 °C
	FlowTemperatureSensor	27,1 °C
	ReturnTemperatureSensor	25,1 °C
	HydraulicSeparatorTemperatureSensor	25,5 °C
	MixerOneCircuitFlowTemperatureSensor	25,5 °C
	WaterPressureSensor	1,70 bar
	PrimaryHeatExchangerLiquidTemperatureSensor	3,9 °C
	CompressorInletTemperatureSensor	6,4 °C
	CompressorInletPressureSensor	4,20 bar
	CompressorOutletTemperatureSensor	37,8 °C
	CompressorOutletPressureSensor	11,90 bar
	SecondaryHeatExchangerLiquidTemperatureSensor	25,3 °C
	CentralHeatingPump	In Betrieb
	DomesticHotWaterPump	Außer Betrieb